Ph429: Nonlinear Dynamics & Chaos

Problem Set 3

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Problem 1

(Problems 3.4.5–10 in Strogatz)
For the following systems, find the values of r at which bifurcations occur, classify the bifurcation as saddle-node, transcritical, or (super- or sub-critical) pitchfork, and sketch the bifurcation diagram of fixed points x^* as a function of r.

If possible, describe how or perform the finding of the normal form for each.

(a)

$$\dot{x} = r - 3x^2$$

Solution

- Find fixed points:

$$\dot{x} = f(x) \quad \Longrightarrow \; f(x^*) = 0$$
$$r - 3x^{*2} = 0$$
$$x^* = \pm\sqrt{\frac{r}{3}}$$

- Vary parameter r and find bifurcation points:
  - $r < 0$: No fixed points
  - $r = 0$: One fixed point at $x^* = 0$
  - $r > 0$: Two fixed points at $x^* = \pm\sqrt{\frac{r}{3}}$

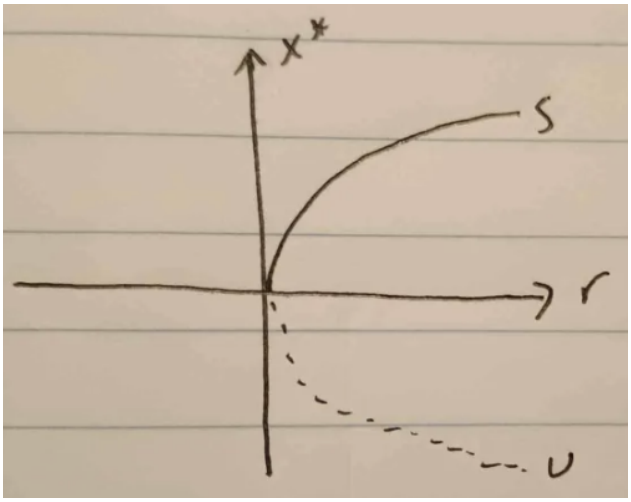
  Since the number of fixed points changes from 0 to 2 at $r = 0$, this is a saddle-node bifurcation with a bifurcation point at $x^* = 0$, $r = 0$.

- Find stability of fixed points:

$$f'(x) = -6x$$

  - $r = 0$: $f'(x^*) = f'(0) = 0$ (inconclusive)
  - $r > 0$:
    - $\rightarrow$ $x^* = +\sqrt{\frac{r}{3}}$: $f'(x^*) < 0$ (stable)
    - $\rightarrow$ $x^* = -\sqrt{\frac{r}{3}}$: $f'(x^*) > 0$ (unstable)

- Bifurcation diagram:



- Normal form: rescale x and r to get $\dot{x} = r - x^2$.

$$3\dot{x} = 3r - 9x^2$$

Let $y = 3x$, $\mu = 3r$

$$\implies \dot{y} = \mu - y^2$$

(b)

$$\dot{x} = rx - \frac{x}{1+x}$$

Solution

- Find fixed points:

$$\dot{x} = f(x) \quad \implies f(x^*) = 0$$
$$x^* \left(r - \frac{1}{1+x^*} \right) = 0$$
$$x^* = 0, \quad x^* = \frac{1}{r} - 1$$

- Vary parameter r and find bifurcation points:
  - $r < 1$: Two fixed points $x^* = 0, \quad x^* = \frac{1}{r} - 1$
  - $r = 1$: One fixed point $x^* = 0$
  - $r > 1$: Two fixed points $x^* = 0, \quad x^* = \frac{1}{r} - 1$
- Find stability of fixed points:

$$f'(x) = r - \frac{(1+x) - x}{(1+x)^2} = r - \frac{1}{(1+x)^2}$$

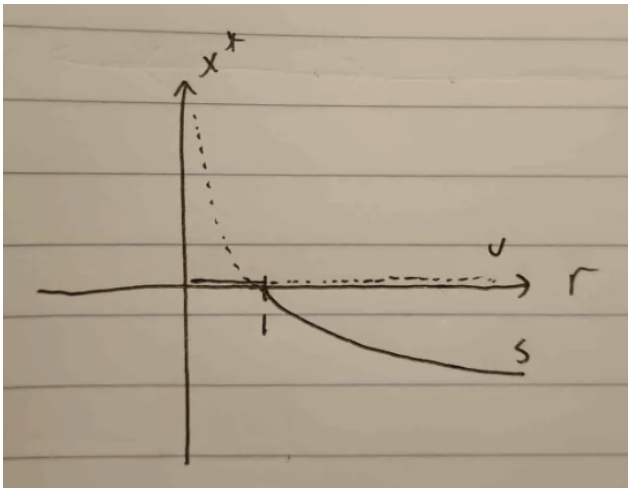
$$f'(0) = r - 1$$

$$f'\left(\frac{1}{r} - 1\right) = r - \frac{1}{\left(1 + \frac{1}{r} - 1\right)^2} = r - \frac{1}{\left(\frac{1}{r}\right)^2} = r - r^2 = r(1-r)$$

- $r < 0$:
  $\rightarrow$ $x^* = 0 : f'(x^*) < 0$ (stable)
  $\rightarrow$ $x^* = \frac{1}{r} - 1$: $f'(x^*) < 0$ (stable)
- $0 < r < 1$:
  $\rightarrow$ $x^* = 0 : f'(x^*) < 0$ (stable)
  $\rightarrow$ $x^* = \frac{1}{r} - 1$: $f'(x^*) > 0$ (unstable)
- $r = 1$:
  $\rightarrow$ $x^* = 0 : f'(x^*) = 0$ (inconclusive)
- $r > 1$:
  $\rightarrow$ $x^* = 0 : f'(x^*) > 0$ (unstable)
  $\rightarrow$ $x^* = \frac{1}{r} - 1 : f'(x^*) < 0$ (stable)
  Note that $r \to 0$ causes $x^* = \frac{1}{r} - 1 \to \infty$. Also, the original equation is not defined at $x = -1$.

  This is a transcritical bifurcation at $x^* = 0$, $r = 1$

- Bifurcation diagram: (RHP)



- Normal form: Using taylor series expansion around the bifurcation point $x^* = 0$

$$\dot{x} = rx - \frac{x}{1+x} \approx rx - x(1 - x + x^2 - \ldots)$$

Neglecting higher order terms, we have

$$\dot{x} \approx rx - x + x^2 = (r-1)x + x^2$$

Let $\mu = r - 1$, we have

$$\dot{x} = \mu x + x^2$$

(c)

$$\dot{x} = 5 - re^{-x^2}$$

Solution

- Find fixed points:

$$\dot{x} = f(x) \implies f(x^*) = 0$$

$$5 - re^{-x^{*2}} = 0 \implies e^{-x^{*2}} = \frac{5}{r} \implies x^{*2} = -\ln\left(\frac{5}{r}\right)$$

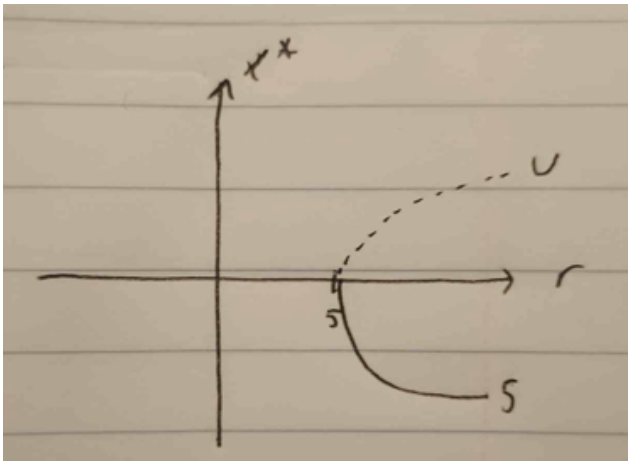
$$x^* = \pm\sqrt{\ln\left(\frac{r}{5}\right)}$$

- Vary parameter r and find bifurcation points:
  For real fixed points, $r/5 \geq 1 \implies r \geq 5$ so that $\ln(r/5) \geq 0$.
  - $r < 5$: No fixed points
  - $r = 5$: One fixed point at $x^* = 0$
  - $r > 5$: Two fixed points at $x^* = \pm\sqrt{\ln\left(\frac{r}{5}\right)}$
- Find stability of fixed points:

$$f'(x) = 2rxe^{-x^2}$$

  - $r = 5$:
    $\rightarrow$ $x^* = 0 : f'(x^*) = 0$ (inconclusive)
  - $r > 5$:
    $\rightarrow$ $x^* = -\sqrt{\ln\left(\frac{r}{5}\right)}$: $f'(x^*) < 0$ (stable)
    $\rightarrow$ $x^* = \sqrt{\ln\left(\frac{r}{5}\right)}$: $f'(x^*) > 0$ (unstable)

  Since we go from 0 fixed points to 2 fixed points at $r = 5$, this is a saddle-node bifurcation at $x^* = 0$, $r = 5$.
- Bifurcation diagram:



- Normal form: $\dot{x} = r + x^2$. Rewriting using a taylor series around the bifurcation point $x^* = 0$ and $r = 5$, we have:

$$\dot{x} = 5 - re^{-x^2} \approx 5 - r\left(1 - x^2 + \frac{x^4}{2} - \dots\right)$$

Neglecting higher order terms, we have

$$\dot{x} \approx 5 - r + rx^2 = (5 - r) + rx^2$$

Let $\mu = 5 - r$, and $r \approx 5$ near the bifurcation:

$$\dot{x} = \mu + 5x^2$$

(d)

$$\dot{x} = rx - \frac{x}{1 + x^2}$$

Solution

- Find fixed points:

$$\dot{x} = f(x) \quad \implies f(x^*) = 0$$
$$x^*\left(r - \frac{1}{1 + x^{*2}}\right) = 0$$
$$x^* = 0, \quad x^* = \pm\sqrt{\frac{1}{r} - 1}$$

- Vary parameter r and find bifurcation points:
  For real fixed points $\frac{1}{r} - 1 \geq 0 \implies 0 < r \leq 1$.
  - $r < 0$: One fixed point $x^* = 0$
  - $0 < r < 1$: Three fixed points $x^* = 0, \quad x^* = \pm\sqrt{\frac{1}{r} - 1}$
  - $r > 1$: One fixed point $x^* = 0$
- Find stability of fixed points:

$$f'(x) = r - \frac{(1 + x^2) - x(2x)}{(1 + x^2)^2} = r - \frac{1 - x^2}{(1 + x^2)^2}$$

$$f'(0) = r - 1$$

$$f'\left(\pm\sqrt{\frac{1}{r} - 1}\right) = r - \frac{1 - \left(\frac{1}{r} - 1\right)}{\left(1 + \frac{1}{r} - 1\right)^2} = r - \frac{2 - \frac{1}{r}}{\left(\frac{1}{r}\right)^2} = r - (2r^2 - r) = 2r - 2r^2 = 2r(1 - r)$$

  - $r < 0$:
    $\rightarrow$ $x^* = 0$: $f'(x^*) < 0$ (stable)
  - $r = 0$:
    $\rightarrow$ $x^* = 0$: $f'(x^*) < 0$ (stable)
  - $0 < r < 1$:
    $\rightarrow$ $x^* = 0$: $f'(x^*) < 0$ (stable)

$\rightarrow$ $x^* = -\sqrt{\frac{1}{r} - 1}$: $f'(x^*) > 0$ (unstable)

$\rightarrow$ $x^* = \sqrt{\frac{1}{r} - 1}$: $f'(x^*) > 0$ (unstable)

- $r = 1$:

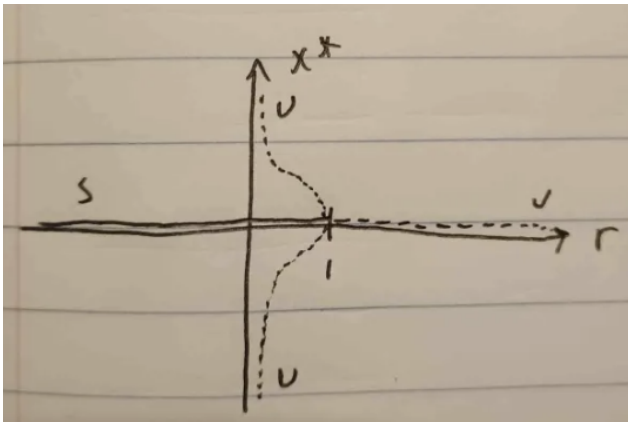
  $\rightarrow$ $x^* = 0$: $f'(x^*) = 0$ (inconclusive)

- $r > 1$:

  $\rightarrow$ $x^* = 0$: $f'(x^*) > 0$ (unstable)

Since we go from 3 fixed points to 1 fixed point where the two symmetrical fixed points are unstable while the middle one is stable, this is a subcritical pitchfork bifurcation at $x^* = 0$, $r = 1$

- Bifurcation diagram:



- Normal form: $\dot{x} = rx + x^3$. Rewriting using a taylor series around the bifurcation point $x^* = 0$ and $r = 1$, we have:

$$\dot{x} = rx - \frac{x}{1 + x^2} \approx rx - x(1 - x^2 + x^4 - \ldots)$$

Neglecting higher order terms, we have

$$\dot{x} \approx rx - x + x^3 = (r - 1)x + x^3$$

Let $\mu = r - 1$, we have

$$\dot{x} = \mu x + x^3$$

(e)

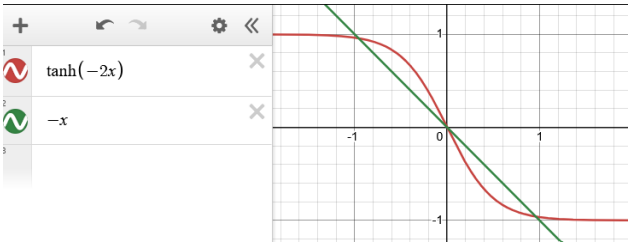
$$\dot{x} = x + \tanh(rx)$$

Solution

- Find fixed points:

$$\dot{x} = f(x) \quad \implies f(x^*) = 0$$
$$x^* + \tanh(rx^*) = 0$$

One fixed point is $x^* = 0$. Graphically, it can be seen that two other fixed points appear for some $r < 0$:



- Since we can't solve for the other fixed points explicitly, I can try finding a bifurcation point based on the assumption that this is some form of subcritical pitchfork bifurcation. Since we start with one fixed point but it becomes three fixed points as r decreases.

  Find r where the fixed point at $x^* = 0$ changes stability:

  $$f'(x) = 1 + r\,sech^2(rx)$$

  $$f'(0) = 1 + r$$

  For $r < -1$, the fixed point at $x^* = 0$ is stable, and for $r > -1$, it is unstable. Therefore, the candidate bifurcation point is at $r = -1$.

- Verify using a taylor series expansion around the candidate bifurcation point $x^* = 0$ and $r = -1$:

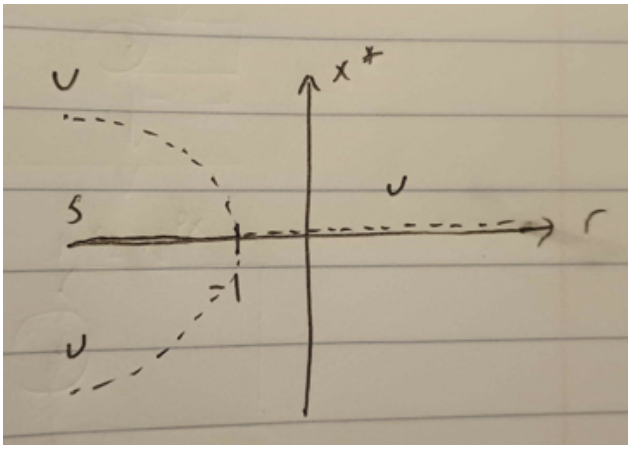
  $$f(x) = x + \tanh(rx) \approx x + \left(rx - \frac{(rx)^3}{3} + \frac{(rx)^5}{5} - \dots\right)$$

  Neglecting higher order terms, we have

  $$f(x) \approx x + rx - \frac{r^3x^3}{3} = (1 + r)x - \frac{r^3}{3}x^3$$

  This has the form of a subcritical pitchfork bifurcation, confirming that the bifurcation point is at $r = -1$.

- Bifurcation diagram:



(f)

$$\dot{x} = rx + \frac{x^3}{1+x^2}$$

Solution

- Find fixed points:

$$\dot{x} = f(x) \quad \implies f(x^*) = 0$$

$$x^*\left(r + \frac{x^{*2}}{1+x^{*2}}\right) = 0$$

$$x^* = 0, \quad x^* = \pm\sqrt{\frac{-r}{1+r}}$$

  For the nonzero fixed points to be real, we need $-r/(1+r) \geq 0 \implies -1 < r < 0$.

- Vary parameter r and find bifurcation points:
  - $r \leq -1$: One fixed point at $x^* = 0$
  - $-1 < r < 0$: Three fixed points at $x^* = 0, \quad x^* = \pm\sqrt{\frac{-r}{1+r}}$
  - $r = 0$: One fixed point at $x^* = 0$
  - $r > 0$: One fixed point at $x^* = 0$
- Find stability of fixed points:

$$f'(x) = r + \frac{3x^2(1+x^2) - x^3(2x)}{(1+x^2)^2} = r + \frac{x^2(3+x^2)}{(1+x^2)^2}$$

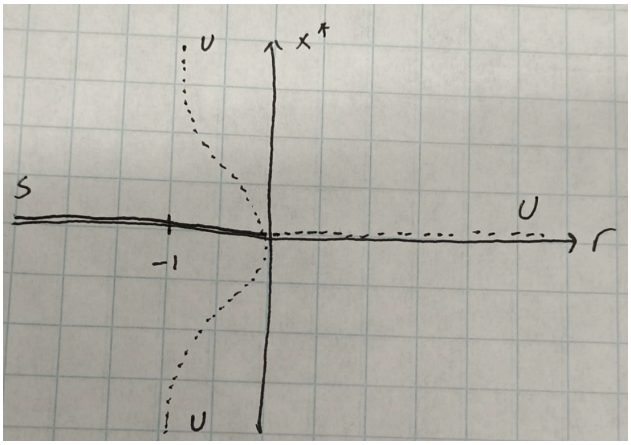
$$f'(0) = r$$

$$f'\left(\pm\sqrt{\frac{-r}{1+r}}\right) = -2r(1+r)$$

  - $r < -1$:
    $\rightarrow$ $x^* = 0$: $f'(x^*) < 0$ (stable)
  - $-1 < r < 0$:
    $\rightarrow$ $x^* = 0$: $f'(x^*) < 0$ (stable)
    $\rightarrow$ $x^* = \pm\sqrt{\frac{-r}{1+r}}$: $f'(x^*) > 0$ (unstable)
  - $r = 0$: One fixed point at $x^* = 0$
    $\rightarrow$ $x^* = 0$: $f'(x^*) = 0$ (inconclusive)
  - $r > 0$: One fixed point at $x^* = 0$
    $\rightarrow$ $x^* = 0$: $f'(x^*) > 0$ (unstable)

  Since we go from 3 fixed points to 1 fixed point, where the two symmetrical fixed points are unstable while the middle one is stable, this is a subcritical pitchfork bifurcation at $x^* = 0$, $r = 0$

- Bifurcation diagram:



- Normal form: $\dot{x} = rx + \frac{x^3}{1+x^2}$. Rewriting using a taylor series around the bifurcation point $x^* = 0$ and $r = 0$, we have:

$$\dot{x} = rx + \frac{x^3}{1+x^2} \approx rx + x^3(1 - x^2 + x^4 - \ldots)$$

Neglecting higher order terms, we have

$$\dot{x} \approx rx + x^3$$

Problem 2

(Problem 3.4.14 in Strogatz)
Subcritical Pitchfork. Consider the system $\dot{x} = rx + x^3 - x^5$, which exhibits a subcritical pitchfork bifurcation (hint: you may want to read about subcritical pitchforks in the book; we did not say much about these in class).

(a) Find algebraic expressions for all the fixed points as r varies.

Solution Find fixed points:

$$\dot{x} = f(x) \quad \implies f(x^*) = 0$$

$$rx^* + x^{*3} - x^{*5} = 0 \quad \implies x^*(r + x^{*2} - x^{*4}) = 0$$

One fixed point at $\boxed{x^* = 0}$. Solve quadratic in terms of $y = x^{*2}$ to find the other fixed points:

$$-y^2 + y + r = 0 \quad \implies y = \frac{-1 \pm \sqrt{1 - 4(-1)(r)}}{-2} = \frac{1 \pm \sqrt{1+4r}}{2}$$

The remaining fixed points are at

$$\boxed{x^* = \pm\sqrt{\frac{1 \pm \sqrt{1+4r}}{2}}}$$

Find number of fixed points as r varies. For the fixed points to be real, r is constrained:

- $x^* = \pm\sqrt{\frac{1+\sqrt{1+4r}}{2}}$

$$1 + \sqrt{1+4r} \geq 0 \implies 1 + 4r \geq 0 \implies r \geq -\frac{1}{4}$$

- $x^* = \pm\sqrt{\frac{1-\sqrt{1+4r}}{2}}$

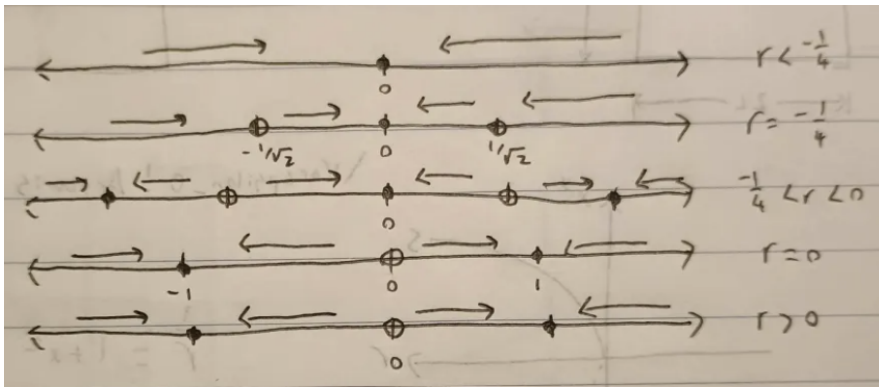
$$1 - \sqrt{1+4r} \geq 0 \implies 1 \geq \sqrt{1+4r} \implies 1 \geq 1 + 4r \implies r \leq 0$$

For

- $r < \frac{-1}{4}$: One fixed point, $x^* = 0$
- $r = \frac{-1}{4}$: Three fixed points, $x^* = 0, \pm\frac{1}{\sqrt{2}}$
- $\frac{-1}{4} < r < 0$: Five fixed points, $x^* = 0, \pm\sqrt{\frac{1+\sqrt{1+4r}}{2}}, \pm\sqrt{\frac{1-\sqrt{1+4r}}{2}}$
- $r = 0$: Three fixed points, $x^* = 0, \pm 1$
- $r > 0$: Three fixed points, $x^* = 0, \pm\sqrt{\frac{1+\sqrt{1+4r}}{2}}$

(b) Sketch the vector fields as r varies. Be sure to indicate all the fixed points and their stability.

Solution



(c) Calculate r_s, the parameter value at which the nonzero fixed points are born in a saddle-node bifurcation.

Solution At a saddle node bifurcation, where $\dot{x} = f(x, r)$:

$$f(x_c, r_c) = 0, \quad \left.\frac{df}{dx}\right|_{(x_c, r_c)} = 0$$

Thus we get the system of equations:

$$f(x_c, r_c) = 0 \implies r_c x_c + x_c^3 - x_c^5 = 0 \implies r_c + x_c^2 - x_c^4 = 0, \quad x_c \neq 0$$

$$\left.\frac{df}{dx}\right|_{(x_c, r_c)} = 0 \implies r_c + 3x_c^2 - 5x_c^4 = 0$$

Subtracting

$$2x_c^2 - 4x_c^4 = 0 \quad \implies 2 - 4x_c^2 = 0, \quad x_c \neq 0$$

$$x_c = \pm\frac{1}{\sqrt{2}}$$

Plugging into $r_c + x_c^2 - x_c^4 = 0$ to find r_c:

$$r_c + \left(\frac{1}{2}\right) - \left(\frac{1}{4}\right) = 0$$

$$\boxed{r_c = r_s = -\frac{1}{4}}$$

Problem 3

What can be said about the stability of a fixed point when the derivative at that fixed point is exactly zero? In other words, what is the stability of x^* when

$$f'(x^*) = \frac{df}{dx}(x^*) = 0\,?$$

Solution For small perturbations around a bifurcation point, the linear stability analysis is inconclusive for $f'(x^*) = 0$. In this case, we would need to either consider the first non-zero higher order derivative of $f(x)$ at x^* or plot the vector field to determine the stability of the fixed point.

Problem 4

Show that the first-order system

$$\dot{x} = x(1 - x^2) - a(1 - e^{-bx})$$

undergoes a transcritical bifurcation at $x = 0$ when the parameters a, b satisfy a certain equation, to be determined. (This equation defines a bifurcation curve in the (a, b) parameter space.) Then find an approximate formula for the fixed point that bifurcates from $x = 0$, assuming that the parameters are close to the bifurcation curve.

Solution I will show that the first order system undergoes a transcritical bifurcation at $x = 0$ by putting it into normal form. The normal form for a transcritical bifurcation is $\dot{x} = \mu x - x^2$. We can rewrite the original system as a Taylor series expansion around the bifurcation point $x^* = 0$:

$$\dot{x} = x(1 - x^2) - a(1 - e^{-bx}) = x - x^3 - a + ae^{-bx}$$

$$\dot{x} \approx x - x^3 - a + a\left(1 - bx + \frac{b^2x^2}{2} - \frac{b^3x^3}{6} + \cdots\right)$$

Neglecting higher order terms, and rewriting:

$$\dot{x} \approx (1 - ab)x + \frac{ab^2}{2}x^2$$

This matches the form of a transcritical bifurcation by rescaling x and defining $\mu = 1 - ab$.

For a bifurcation to occur at $x = 0$, we have the condition that $\dot{x} = f(x) = 0$ at $x = 0$. That is, the bifurcation point must also be a fixed point.

$$(1 - ab)x + \frac{ab^2}{2}x^2 = 0 \quad \text{at } x = 0$$

It also has to satisfy the condition that the derivative at the bifurcation point is zero such that the stability of the fixed point changes at the bifurcation point.

$$f'(0) = 1 - ab = 0 \quad \Longrightarrow \quad \boxed{ab = 1}$$

Finally, solving for the fixed points:

$$f(x*) = 0 \quad \Longrightarrow (1 - ab)x^* + \frac{ab^2}{2}x^{*2} = 0$$

Fixed point at $\boxed{x^* = 0}$ and another fixed point at

$$(1 - ab) + \frac{ab^2}{2}x^* = 0$$

$$\boxed{x^* = \frac{2(ab - 1)}{ab^2}}$$